

National Exam of Higher National Diploma-New program – 2020 Session

Spécialty/option : SWE-DBM

Paper : Information System

Duration : 3 hours



Credit : 6

Instructions: Follow the instructions below carefully

- This paper contains three sections; Section A (40%), Section B (40%) and Section C (20%). Answer all questions in each section.
- The respective mark(s) of each question is indicated.
- It is a closed book exam and pre-prepared material is not authorized.
- Should any student encounter any difficulty, only the examiner(s) in the examination hall should be contacted

SECTION A: SYSTEM ARCHITECTURE. (40marks)

1. What is an Object server in client server environment? (4mks)
2. What are the two broad classes of middleware in client server environment? (4mks)
3. What are the five major technologies that can be used to create Client/Server applications in client server environment? (4mks)
4. What is public cloud and private cloud? (4mks)
5. What are the essential characteristics of cloud computing? (4mks)
6. How many types of deployment models cloud? (4mks)
7. Describe about the different Guided Medias. (4mks)
8. What do you mean by wireless communication? (4mks)
9. List two (02) advantages and two (02) disadvantages of fiber optics cable. (4mks)
10. List three (03) factors on which data rate depends. (4mks)

SECTION B: OPERATING SYSTEMS. (40marks)

1. Explain the main purpose of an operating system? (5mks)
2. What are the advantages of a multiprocessor system? (5mks)
3. Describe the objective of multiprogramming. (5mks)
4. Give some benefits of multithreaded programming. (5mks)
5. i) What is a file ? (1mk)
ii) What are the Operations performed on files or directories (4mks)
6. What Is the Google Android SDK? (5mks)
7. List the four (04) key components of Android Architecture. (5mks)
8. What is the importance of having an emulator within the Android environment?(5mks)

SECTION C: PROJECT MANAGEMENT AND LEGAL REGULATIONS. (20marks)

1. What is the difference between agile project management and agile software development? **(5mks)**
2. How important is "Project Methodology" as a parameter for software cost estimation? **(5mks)**
3. Can agile development be applied in traditional product development? **(5mks)**
4. Explain what is Copyright. **(5mks)**